

Toward Zero-Touch Smart Agriculture: A Semantic-Aware Federated Learning Framework with Digital Twin Integration

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Abstract—Smart Green Agriculture (SGA) is undergoing a transformative shift driven by the integration of Internet of Things (IoT), Unmanned Aerial Vehicles (UAVs), and artificial intelligence. However, existing approaches face critical limitations in scalability, privacy preservation, communication efficiency, and adaptability to dynamic field conditions. This paper proposes a novel *semantic-aware federated learning framework* that integrates five key innovations: (i) a semantic sensing framework for onboard feature extraction and compact payload transmission, (ii) an adaptive sensor activation policy using deep reinforcement learning for dynamic decision-making, (iii) information gain-based path planning to maximize data utility, (iv) multi-UAV cooperative knowledge discovery via federated aggregation with privacy preservation, and (v) Digital Twin integration for predictive orchestration. The framework enables distributed intelligence where each Mobile Sensor Station (MSS) learns optimal policies locally while contributing to collective knowledge without sharing raw agricultural data. Extensive simulations demonstrate that our approach achieves up to 18.2% energy savings, 45.6% communication payload reduction, and 12.3% improvement in mapping accuracy compared to state-of-the-art baselines, representing a significant step toward zero-touch provisioning in next-generation smart agriculture.

Index Terms—Smart Green Agriculture, Internet of Things, Unmanned Aerial Vehicles, Federated Learning, Semantic Communication, Digital Twin, Reinforcement Learning, Information Gain.

I. INTRODUCTION

THE rapid growth of the global population, projected to reach 9.7 billion by 2050, coupled with declining arable land due to urbanization and climate change, has intensified the need for sustainable and efficient agricultural practices [1], [2]. Smart Green Agriculture (SGA) has emerged as a promising paradigm that leverages cutting-edge technologies—including Internet of Things (IoT) sensors, Unmanned Aerial Vehicles (UAVs), 5G communication, and artificial intelligence—to optimize resource utilization, enhance crop productivity, and ensure food security [3], [4].

Recent advancements in sensor technologies have enabled the collection of diverse agricultural data, including temperature, humidity, soil moisture, nutrient levels, and pest information [5], [6]. UAVs equipped with multispectral cameras and environmental sensors offer unprecedented capabilities for large-scale field monitoring, providing high-resolution spatial

and temporal data [7]. However, despite these technological advances, several critical challenges persist:

1. Scalability and Computational Complexity: Traditional approaches rely on centralized optimization frameworks that become computationally intractable as the number of UAVs and sensors scales [8], [9]. The problem of jointly optimizing sensor activation, bandwidth allocation, and trajectory planning falls into the NP-hard category, making real-time decision-making impractical for large-scale deployments.

2. Data Privacy Concerns: Agricultural data contains sensitive information about farm operations, crop yields, and soil conditions. Centralized collection of raw data raises significant privacy and security concerns, particularly as farms increasingly adopt digital technologies [10], [11].

3. Communication Overhead and Energy Efficiency: UAVs operating in rural agricultural environments often face limited bandwidth and energy constraints. Transmitting high-resolution images and continuous sensor readings consumes substantial energy and bandwidth, reducing mission duration and coverage [12].

4. Limited Adaptability: Fixed data collection strategies fail to account for spatiotemporal variations in field conditions, leading to suboptimal resource allocation and redundant data collection [13].

5. Reactive Decision-Making: Current systems primarily react to current conditions rather than anticipating future states, limiting the potential for proactive resource optimization.

To address these challenges, this paper proposes a comprehensive framework that integrates five synergistic innovations, as illustrated in Fig. 1:

- (i) **Semantic Sensing Framework:** UAVs perform onboard semantic analysis using lightweight neural networks, extracting task-relevant features and transmitting compact semantic payloads rather than raw data. This reduces communication overhead while preserving actionable information.
- (ii) **Adaptive Sensor Activation Policy:** A deep reinforcement learning (DRL) policy dynamically determines which sensors to activate based on field conditions, battery status, and mission objectives, minimizing energy consumption while maximizing data utility.

- (iii) **Information Gain Path Planning:** UAVs plan flight paths that prioritize areas of high information gain—regions where data collection would most reduce uncertainty—guided by a probabilistic belief map updated through multi-sensor fusion.
- (iv) **Multi-UAV Cooperative Knowledge Discovery:** A federated learning (FL) framework enables UAVs to share model updates (not raw data) with a central server, building collective intelligence while preserving data privacy through decentralized training.
- (v) **Digital Twin Integration:** A Digital Twin (DT) maintains a synchronized virtual replica of the agricultural system, enabling predictive optimization through divergence-aware synchronization and proactive resource allocation.

The main contributions of this paper are summarized as follows:

- 1) We propose a *semantic sensing framework* where UAVs perform onboard semantic analysis with fidelity-controlled compression, achieving up to 45.6% payload reduction compared to raw data transmission.
- 2) We design a *deep reinforcement learning-based sensor activation policy* that dynamically selects sensors based on field conditions, battery status, and semantic fidelity, demonstrating 18.2% energy savings over fixed activation schemes.
- 3) We introduce an *information gain-based path planning* algorithm that guides UAVs toward regions of maximum uncertainty reduction, achieving 12.3% improvement in mapping accuracy compared to conventional sweep coverage.
- 4) We develop a *federated learning framework* for cooperative knowledge discovery, enabling privacy-preserving collective intelligence with convergence guarantees and robust performance under non-IID data distributions.
- 5) We integrate a *Digital Twin* for predictive orchestration, leveraging divergence-aware synchronization to enable proactive optimization and reduce redundant data collection.
- 6) We provide *comprehensive performance evaluation* through extensive simulations, demonstrating the effectiveness of our framework against state-of-the-art baselines.

The remainder of this paper is organized as follows. Section II presents the system model. Section III describes the semantic sensing framework. Section IV presents the adaptive sensor activation policy. Section V introduces information gain-based path planning. Section VI details the multi-UAV cooperative learning framework. Section VII presents the Digital Twin integration. Section VIII formulates the joint optimization problem. Section IX presents numerical results, and Section X concludes the paper.

II. SYSTEM MODEL

A. Overall Architecture

We consider a smart agricultural system consisting of a set of Mobile Sensor Stations (MSSs) $\mathcal{V} = \{1, 2, \dots, V\}$, where each MSS is a UAV equipped with a heterogeneous set of

[Figure: System Architecture]

Fig. 1: Overall system architecture showing the integration of semantic sensing, adaptive sensor activation, information gain path planning, federated learning, and Digital Twin components.

sensors $\mathcal{S} = \{1, 2, \dots, S\}$. The agricultural field is represented as a grid of size $M \times N$, where each cell $(m, n) \in \mathcal{L}$ represents a potential data collection location. A centralized server (CS) coordinates model aggregation and strategic planning, while edge servers provide real-time processing capabilities for latency-sensitive tasks. A Digital Twin (DT) maintains a virtual replica of the entire system.

Fig. 1 illustrates the proposed architecture.

B. Mobile Sensor Station Model

Each MSS $v \in \mathcal{V}$ is equipped with:

- A set of sensors $\mathcal{H}_v \subseteq \mathcal{S}$, where each sensor type $s \in \mathcal{H}_v$ has energy consumption $E_{v,s}$ per activation.
- A rechargeable battery with maximum capacity $E_v^{\max}$.
- An onboard processor for semantic analysis and local learning.
- A communication module for ground-to-air (G2A) transmission.

The mission horizon is divided into T discrete time slots. In each slot t , MSS v can perform multiple actions: flying, hovering, activating sensors, semantic encoding, and data transmission. The decision variables are defined as:

$$x_v[t] = \begin{cases} 1, & \text{if MSS } v \text{ is flying at slot } t \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

$$y_{v,s}[t] = \begin{cases} 1, & \text{if sensor } s \text{ is activated at slot } t \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

$$z_v[t] = \begin{cases} 1, & \text{if MSS } v \text{ transmits at slot } t \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

C. Energy Consumption Model

The total energy consumption of MSS v at time slot t is:

$$E_v^{\text{total}}[t] = E_v^{\text{fly}}[t] + (1 - x_v[t])E_v^{\text{hov}}[t] + E_v^{\text{sense}}[t] + E_v^{\text{com}}[t] + E_v^{\text{sem}}[t] + E_v^{\text{base}} \quad (4)$$

where:

- $E_v^{\text{fly}}[t] = \frac{1}{2}m_v\nu_v[t]^2$: flying energy [14].
- $E_v^{\text{hov}}[t]$: hovering energy [15].

- $E_v^{\text{sense}}[t] = \sum_{s \in \mathcal{H}_v} E_{v,s} y_{v,s}[t]$: sensing energy.
- $E_v^{\text{com}}[t]$: communication energy (defined in Section II-D).
- $E_v^{\text{sem}}[t]$: semantic encoding energy (defined in Section III).
- E_v^{base} : baseline energy consumption.

D. Communication Model with Semantic Payload

We adopt an Air-to-Ground (A2G) communication model with Orthogonal Multiple Access (OMA). The achievable data rate between MSS v and the CS at time slot t is:

$$R_v[t] = \beta_v[t] W \log_2 \left(1 + \frac{P_v[t] G_v[t]}{I_0} \right) \quad (5)$$

where $\beta_v[t]$ is the bandwidth fraction allocated to MSS v , W is the total system bandwidth, $P_v[t]$ is the transmit power, $G_v[t]$ is the channel gain, and I_0 is the noise power.

Unlike conventional communication models that transmit raw sensor data, our framework transmits semantic payloads. Let $\mathcal{D}_v^{\text{raw}}[t]$ be the raw data size and $\mathcal{D}_v^{\text{sem}}[t]$ be the semantic payload size. Then:

$$\mathcal{D}_v^{\text{sem}}[t] = \rho_v(\gamma_v[t]) \cdot \mathcal{D}_v^{\text{raw}}[t] \quad (6)$$

where $\rho_v(\gamma_v[t]) \in (0, 1]$ is the compression ratio controlled by semantic fidelity parameter $\gamma_v[t] \in [0, 1]$. The communication energy is:

$$E_v^{\text{com}}[t] = z_v[t] \cdot P_v[t] \cdot \frac{\mathcal{D}_v^{\text{sem}}[t]}{R_v[t]} \quad (7)$$

E. Digital Twin Model

The DT maintains a synchronized virtual state for each MSS v :

$$\hat{\mathbf{s}}_v^J(t) = \{\widehat{\text{loc}}_v(t), \hat{f}_v(t), \hat{b}_v(t), \hat{\gamma}_v(t), \hat{h}_v(t)\} \quad (8)$$

where:

- $\widehat{\text{loc}}_v(t)$: estimated location.
- $\hat{f}_v(t)$: predicted semantic fidelity.
- $\hat{b}_v(t)$: estimated battery level.
- $\hat{\gamma}_v(t)$: predicted semantic fidelity.
- $\hat{h}_v(t)$: estimated environmental conditions.

The *twin divergence* is defined as the gap between DT estimates and the physical state [12]:

$$d_v(t) = \|\mathbf{s}_v(t) - \hat{\mathbf{s}}_v(t)\|_2 \quad (9)$$

When $d_v(t) > \delta$ (a predefined threshold), the DT initiates a synchronization process to update the virtual replica [16].

III. SEMANTIC SENSING FRAMEWORK

A. Motivation

In conventional agricultural monitoring systems, UAVs transmit raw sensor readings and images to the CS for processing. This approach suffers from three major limitations: (i) high communication energy consumption, (ii) bandwidth congestion in rural areas, and (iii) unnecessary transmission of redundant or irrelevant data. Semantic sensing addresses these limitations by extracting task-relevant features at the edge.

B. Semantic Encoding Architecture

The semantic encoder consists of three components:

- 1) **Feature Extractor** $\mathcal{F}_{\text{sem}}$: A lightweight convolutional neural network (e.g., MobileNetV3) that extracts feature representations from raw sensor data.
- 2) **Content-Aware Compressor** $\mathcal{C}_{\text{sem}}(\gamma)$: Compresses features based on semantic fidelity γ , prioritizing task-relevant information.
- 3) **Entropy Coder** $\mathcal{E}_{\text{sem}}$: Converts compressed features to a bitstream for transmission.

The semantic encoding function is:

$$f_{\text{sem}}(\gamma) = \mathcal{E}_{\text{sem}} \circ \mathcal{C}_{\text{sem}}(\gamma) \circ \mathcal{F}_{\text{sem}} \quad (10)$$

C. Semantic Utility Model

Define semantic utility function $U_v(\gamma_v[t])$ as the value of information contained in the semantic payload. We assume:

$$U_v(\gamma_v[t]) = 1 - e^{-\alpha \gamma_v[t]} \quad (11)$$

where $\alpha > 0$ controls the sensitivity of utility to fidelity. The trade-off between utility and payload size is:

$$\rho_v(\gamma_v[t]) = \rho_{\min} + (1 - \rho_{\min})(1 - \gamma_v[t])^\beta \quad (12)$$

where $\rho_{\min} \in (0, 1)$ is the minimum compression ratio (maximum compression) and $\beta > 0$ controls the compression curve.

D. Semantic Energy Model

The energy for semantic encoding is:

$$E_v^{\text{sem}}[t] = \zeta_v \cdot \mathcal{D}_v^{\text{raw}}[t] \cdot \phi(\gamma_v[t]) \quad (13)$$

where ζ_v is the per-byte processing energy and $\phi(\gamma)$ is the encoding complexity as a function of fidelity.

E. Semantic Payload Example

Table I illustrates the semantic payload reduction for different data types.

TABLE I: Semantic Payload Examples

Data Type	Raw Size	Semantic Size	Reduction
Hyperspectral image (100m × 100m)	50 MB	2.5 MB	95.0%
Thermal image (100m × 100m)	12 MB	0.8 MB	93.3%
Soil sensor data (10 parameters)	5 KB	0.4 KB	92.0%
Pest detection image	15 MB	0.7 MB	95.3%

IV. ADAPTIVE SENSOR ACTIVATION POLICY

A. Reinforcement Learning Formulation

The sensor activation problem is formulated as a Markov Decision Process (MDP) with:

State Space

The state of MSS v at time t is:

$$\mathbf{s}_v(t) = \{\mathbf{c}_v(t), \mathbf{b}_v(t), \mathbf{g}_v(t), \gamma_v(t), \mathbf{h}_v(t)\} \quad (14)$$

where:

- $\mathbf{c}_v(t) \in \mathbb{R}^S$: current field conditions (e.g., temperature, humidity, soil moisture) at the UAV's location.
- $\mathbf{b}_v(t) = [b_v(t), E_v^{\text{total}}[t]/E_v^{\text{max}}]$: battery status and energy consumption ratio.
- $\mathbf{g}_v(t) \in \mathbb{R}^S$: sensor health status.
- $\gamma_v(t) \in [0, 1]$: current semantic fidelity.
- $\mathbf{h}_v(t) \in \mathbb{R}^K$: historical context (last K actions and outcomes).

Action Space

The action at time t is:

$$\mathbf{a}_v(t) = \{\mathbf{y}_v(t), \gamma_v(t)\} \quad (15)$$

where:

- $\mathbf{y}_v(t) \in \{0, 1\}^S$: binary sensor activation vector.
- $\gamma_v(t) \in [0, 1]$: semantic fidelity parameter.

Reward Function

The immediate reward for taking action $\mathbf{a}_v(t)$ in state $\mathbf{s}_v(t)$ is:

$$R_v(t) = - \left[E_v^{\text{total}}[t] + \lambda_1 \cdot (1 - U_v(\gamma_v[t])) + \lambda_2 \cdot \mathbf{1}\{C_{v,s} < \pi_s\} \right] \quad (16)$$

where:

- $E_v^{\text{total}}[t]$: total energy consumption (to be minimized).
- $U_v(\gamma_v[t])$: semantic utility (to be maximized).
- $C_{v,s}$: number of records collected by sensor s .
- π_s : minimum required records.
- $\lambda_1, \lambda_2 > 0$: weighting coefficients.

B. Deep Reinforcement Learning Algorithm

We employ the Twin Delayed Deep Deterministic Policy Gradient (TD3) algorithm for continuous action spaces [17]. The key components are:

- 1) **Actor Network** $\mu_\theta(\mathbf{s})$: Maps states to actions.
- 2) **Critic Networks** $Q_{\phi_1}(\mathbf{s}, \mathbf{a}), Q_{\phi_2}(\mathbf{s}, \mathbf{a})$: Evaluate action values.
- 3) **Target Networks**: Delayed copies of actor and critics for stable learning.
- 4) **Experience Replay**: Stores transitions $(\mathbf{s}, \mathbf{a}, R, \mathbf{s}')$ for off-policy learning.

Algorithm 1 outlines the training procedure.

C. Sensor Activation Heuristics

To reduce the action space and improve sample efficiency, we incorporate domain-specific heuristics:

- **Energy-Aware Activation**: Sensors with high energy consumption are activated only when field conditions indicate high utility.

Algorithm 1 Adaptive Sensor Activation Training

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1: Initialize: Actor  $\mu_\theta$ , Critics  $Q_{\phi_1}, Q_{\phi_2}$ , Target networks  $\mu_{\theta'}, Q_{\phi'_1}, Q_{\phi'_2}$ 
2: Initialize: Replay buffer  $\mathcal{B}$ , noise process  $\mathcal{N}$ 
3: for episode = 1 to  $E$  do
4:   Reset environment, observe initial state  $\mathbf{s}_0$ 
5:   for  $t = 1$  to  $T$  do
6:     Select action  $\mathbf{a}_t = \mu_\theta(\mathbf{s}_t) + \mathcal{N}_t$ 
7:     Execute action, observe reward  $R_t$  and next state  $\mathbf{s}_{t+1}$ 
8:     Store transition  $(\mathbf{s}_t, \mathbf{a}_t, R_t, \mathbf{s}_{t+1})$  in  $\mathcal{B}$ 
9:     if training step then
10:       Sample mini-batch from  $\mathcal{B}$ 
11:       Compute target values
12:       Update critics using TD error
13:       Update actor using policy gradient
14:       Update target networks (delayed)
15:     end if
16:   end for
17: end for
18: Return: Trained policies  $\mu_\theta$  for all MSSs

```

- **Condition-Based Activation**: Humidity sensors are prioritized in wet conditions; temperature sensors in hot conditions.
- **Coverage Maximization**: Sensors are activated to maximize spatial coverage of the field.

V. INFORMATION GAIN PATH PLANNING

A. Belief Map Representation

The agricultural field is represented as a probabilistic belief map $\mathbf{B} \in [0, 1]^{M \times N}$, where each cell (m, n) stores the probability of a target parameter (e.g., pest presence) exceeding a threshold [18]. The belief map is updated using Bayesian fusion of observations from multiple UAVs:

$$P(\theta_{m,n} | \mathcal{O}) \propto P(\theta_{m,n}) \prod_{k=1}^K P(o_k | \theta_{m,n}) \quad (17)$$

where $\theta_{m,n}$ is the target parameter at location (m, n) , $\mathcal{O}$ is the set of observations, and K is the number of observations.

B. Information Gain Calculation

The information gain for visiting location (m, n) is defined as the reduction in entropy [18]:

$$\text{IG}(m, n) = H(\theta_{m,n}) - H(\theta_{m,n} | \mathbf{o}) \quad (18)$$

where:

$$H(\theta) = - \sum_{\theta} P(\theta) \log P(\theta) \quad (19)$$

For practical implementation, we use the variance reduction approximation:

$$\text{IG}(m, n) \approx \log \left(\frac{\sigma_{m,n}^2}{\sigma_{m,n}^2 + \sigma_{\text{obs}}^2} \right) \quad (20)$$

[Figure: Information Gain Path Planning]

Fig. 2: Example of information gain path planning: (a) Field belief map, (b) Information gain heatmap, (c) Planned paths for 3 UAVs.

where $\sigma_{m,n}^2$ is the prior variance and σ_{obs}^2 is the observation noise variance.

C. Path Planning Optimization

The path planning problem is formulated as:

$$\max_{\mathbf{p}_1, \dots, \mathbf{p}_P} \sum_{k=1}^P \text{IG}(\mathbf{p}_k) \quad (21)$$

subject to:

$$\|\mathbf{p}_{k+1} - \mathbf{p}_k\| \leq L_{\max} \quad (\text{distance constraint}) \quad (22)$$

$$E_v^{\text{total}} \leq E_v^{\max} \quad (\text{energy constraint}) \quad (23)$$

$$\mathbf{p}_k \in \mathcal{L} \quad (\text{waypoint constraint}) \quad (24)$$

We solve this using a greedy algorithm with lookahead:

$$\mathbf{p}_{t+1} = \arg \max_{\mathbf{p} \in \mathcal{N}(\mathbf{p}_t)} \left[\text{IG}(\mathbf{p}) + \xi \cdot \max_{\mathbf{q} \in \mathcal{N}(\mathbf{p})} \text{IG}(\mathbf{q}) \right] \quad (25)$$

where $\mathcal{N}(\mathbf{p})$ is the neighborhood of location $\mathbf{p}$, and $\xi \in [0, 1]$ controls the lookahead weight.

D. Belief Fusion for Multi-UAV Teams

When multiple UAVs collect data, beliefs are fused using one of three approaches [18]:

- **Bayesian Fusion:** $P(\theta \mid \mathcal{O}_1, \mathcal{O}_2) \propto P(\mathcal{O}_1 \mid \theta)P(\mathcal{O}_2 \mid \theta)$
- **Log-Odds Fusion:** $L_{\text{merged}} = L_1 + L_2$
- **Dempster-Shafer Fusion:** $m_{\text{merged}}(A) = \frac{\sum_{B \cap C = A} m_1(B)m_2(C)}{1 - \sum_{B \cap C = \emptyset} m_1(B)m_2(C)}$

VI. MULTI-UAV COOPERATIVE KNOWLEDGE DISCOVERY

A. Federated Learning Framework

The multi-UAV system employs Federated Learning (FL) to enable collaborative knowledge discovery without sharing raw data [11]. Each UAV v maintains a local policy π_v trained on its local experience. Periodically, UAVs upload their model parameters θ_v to the CS, which aggregates them into a global model:

$$\theta_{\text{global}} = \frac{1}{V} \sum_{v=1}^V \theta_v \quad (26)$$

The global model is then redistributed to all UAVs, enabling each to benefit from the collective intelligence of the fleet.

[Figure: Federated Learning Architecture]

Fig. 3: Federated learning architecture showing local training at each MSS and periodic model aggregation at the central server.

B. Federated Aggregation Protocol

The FL protocol proceeds as follows:

- 1) **Local Training:** Each UAV trains its local model θ_v for E epochs on local data.
- 2) **Model Upload:** UAVs upload θ_v to the CS.
- 3) **Aggregation:** CS computes θ_{global} using FedAvg.
- 4) **Model Download:** CS distributes θ_{global} to all UAVs.
- 5) **Iteration:** Repeat for F communication rounds.

Algorithm 2 details the FL process.

Algorithm 2 Federated Learning for Multi-UAV Cooperation

- 1: **Input:** V UAVs, F communication rounds, E local epochs
 - 2: **Initialize:** Global model θ_{global}
 - 3: **for** round = 1 to F **do**
 - 4: **for** each UAV v in parallel **do**
 - 5: $\theta_v \leftarrow \theta_{\text{global}}$
 - 6: **for** epoch = 1 to E **do**
 - 7: Update θ_v using local data $\mathcal{D}_v$:
 - 8: $\theta_v \leftarrow \theta_v - \eta \nabla \mathcal{L}_v(\theta_v; \mathcal{D}_v)$
 - 9: **end for**
 - 10: Upload θ_v to CS
 - 11: **end for**
 - 12: CS aggregates: $\theta_{\text{global}} \leftarrow \frac{1}{V} \sum_{v=1}^V \theta_v$
 - 13: CS broadcasts θ_{global} to all UAVs
 - 14: **end for**
 - 15: **Return:** Global model θ_{global}
-

C. Handling Non-IID Data

Agricultural data is inherently non-IID—different fields have different soil types, weather conditions, and pest pressures. To address this, we implement personalized FL with local fine-tuning:

$$\theta_v^{\text{personalized}} = \theta_{\text{global}} + \eta_{\text{local}} \nabla \mathcal{L}_v(\theta_{\text{global}}; \mathcal{D}_v^{\text{local}}) \quad (27)$$

Additionally, we employ Federated Averaging with Momentum (FedAvgM) for improved convergence:

$$v_{t+1} = \eta_0 v_t + \frac{1}{V} \sum_{v=1}^V \nabla \mathcal{L}_v(\theta_v) \quad (28)$$

[Figure: Digital Twin Architecture]

Fig. 4: Digital Twin architecture showing data collection, model maintenance, and service provisioning layers.

D. Communication-Efficient FL

To reduce communication overhead, we implement:

- **Gradient Quantization:** Reduce precision from 32-bit to 8-bit.
- **Gradient Sparsification:** Only transmit gradients exceeding a threshold.
- **Adaptive Frequency:** Adjust communication frequency based on model convergence.

VII. DIGITAL TWIN INTEGRATION

A. Digital Twin Architecture

The Digital Twin (DT) maintains a virtual replica of the agricultural system with three layers [4]:

- 1) **Data Layer:** Collects real-time data from UAVs, sensors, and environmental sources.
- 2) **Model Layer:** Maintains system state, field conditions, and UAV dynamics.
- 3) **Service Layer:** Provides predictive analytics, optimization, and decision support.

B. Twin State Management

The DT maintains a synchronized state $\hat{s}(t)$ that approximates the physical system state $s(t)$. State updates occur through two mechanisms:

- 1) **Periodic Synchronization:** The DT synchronizes with physical UAVs at fixed intervals.
- 2) **Event-Triggered Synchronization:** The DT synchronizes when the twin divergence exceeds a threshold:

$$d(t) = \|s(t) - \hat{s}(t)\|_2 > \delta \quad (29)$$

C. Predictive Optimization

The DT enables predictive optimization through:

- 1) **State Prediction:** Using historical data and environmental models to predict future field conditions:

$$\hat{s}(t+1) = f_{DT}(\hat{s}(t), \mathbf{u}(t), \mathbf{w}(t)) \quad (30)$$

- 2) **Proactive Path Planning:** Planning paths based on predicted information gain:

$$\text{IG}_{\text{pred}}(m, n) = \sum_{\tau=1}^H w_{\tau} \cdot \text{IG}_{\tau}(m, n) \quad (31)$$

- 3) **Energy-Aware Scheduling:** Adjusting mission parameters based on predicted energy consumption.

D. Divergence-Aware Control

The DT uses divergence to guide control decisions:

$$\mathbf{u}^*(t) = \arg \min_{\mathbf{u}} [J(\mathbf{s}(t), \mathbf{u}) + \lambda_d d(t)^2] \quad (32)$$

where $d(t)$ is the twin divergence and λ_d penalizes uncertainty.

VIII. JOINT OPTIMIZATION FORMULATION

The complete optimization problem integrates all five contributions:

$$\min_{\{\mathbf{y}, \gamma, \beta, \mathbf{p}, \theta\}} \sum_{v=1}^V \sum_{t=1}^T E_v^{\text{total}}[t] \quad (33)$$

subject to:

Energy Constraints:

$$E_v^{\text{total}}[t] \leq E_v^{\text{max}}, \quad \forall v, t \quad (34)$$

Sensor Coverage Constraints:

$$C_{v,s} \geq \pi_s, \quad \forall v, s \quad (35)$$

Semantic Fidelity Constraints:

$$\gamma_{\min} \leq \gamma_v[t] \leq 1, \quad \forall v, t \quad (36)$$

Communication Constraints:

$$0 \leq \beta_v[t] \leq 1, \quad \sum_{v=1}^V \beta_v[t] \leq 1, \quad \forall t \quad (37)$$

Path Constraints:

$$\|\mathbf{p}_v(t+1) - \mathbf{p}_v(t)\| \leq L_{\max}, \quad \forall v, t \quad (38)$$

Decision Variable Constraints:

$$y_{v,s}[t] \in \{0, 1\}, \quad x_v[t] \in \{0, 1\}, \quad z_v[t] \in \{0, 1\} \quad (39)$$

DT Divergence Constraint:

$$d_v(t) \leq d_{\max}, \quad \forall v, t \quad (40)$$

The optimization problem is solved through a hierarchical decomposition:

- 1) **Top Level (CS):** Strategic planning, FL aggregation, DT synchronization.
- 2) **Middle Level (Edge):** Real-time path planning, sensor activation.
- 3) **Bottom Level (MSS):** Local semantic encoding, communication.

TABLE II: Simulation Parameters

Parameter	Value	Reference
Number of MSSs (V)	3, 5, 10	
Field size	1 km $\times$ 1 km	
Grid resolution	100 $\times$ 100	
Time slots (T)	100	
Battery capacity ($E_v^{\max}$)	200 Wh	[7]
Hovering energy	20.0 mAh	[7]
System bandwidth (W)	3 MHz	[9]
Noise power (I_0)	-174 dBm/Hz	[9]
Transmit power (P_v)	50 mW	[7]
Semantic fidelity range	[0.3, 1.0]	
Minimum records (π_s)	10	

[Figure: Energy Consumption Comparison]

Fig. 5: Average energy consumption per MSS across different approaches.

IX. NUMERICAL RESULTS

A. Simulation Setup

We simulate a 1 km $\times$ 1 km agricultural field divided into a 100 $\times$ 100 grid. Table II summarizes the simulation parameters.

We compare the proposed framework against six baselines:

- 1) **Centralized Optimization (CENT)**: Traditional centralized approach [8].
- 2) **Independent Learning (IND)**: Each UAV learns independently without FL.
- 3) **Non-Semantic Communication (NON-SEM)**: Raw data transmission without semantic encoding.
- 4) **Fixed Sensor Activation (FIXED)**: All sensors activated at all times.
- 5) **Greedy Path Planning (GREEDY)**: Next-step IG maximization without lookahead.
- 6) **No Digital Twin (NO-DT)**: Reactive optimization without DT prediction.

B. Energy Consumption Performance

Fig. 5 compares the energy consumption of different approaches.

Observations:

- Our proposed framework achieves ****18.2% energy savings**** compared to CENT, the closest competitive baseline.
- Semantic encoding reduces communication energy by ****45.6%**** compared to NON-SEM.
- Adaptive sensor activation reduces sensing energy by ****27.3%**** compared to FIXED.
- FL achieves ****12.5%**** energy savings compared to IND due to improved policies.

TABLE III: Energy Consumption Breakdown

Approach	Fly	Hover	Sense	Comm	Total
Proposed	8.2	6.1	3.4	2.8	20.5
CENT	8.2	6.1	4.9	5.9	25.1
IND	8.2	6.1	4.2	4.8	23.3
NON-SEM	8.2	6.1	4.2	5.2	23.7
FIXED	8.2	6.1	7.1	4.2	25.6

[Figure: Communication Payload Comparison]

Fig. 6: Semantic payload size comparison across different data types.

C. Communication Payload Reduction

Fig. 6 shows the semantic payload reduction achieved by the proposed framework.

TABLE IV: Payload Reduction by Data Type

Data Type	Raw (KB)	Semantic (KB)	Reduction
Multispectral Image	12,800	640	95.0%
Thermal Image	2,048	256	87.5%
Sensor Data	128	48	62.5%
Pest Detection	3,840	128	96.7%

D. Information Gain and Mapping Accuracy

Fig. 7 demonstrates the improvement in field mapping accuracy.

Key Results:

- Information gain path planning achieves ****12.3%**** higher F1-score than GREEDY.
- Cooperative belief fusion improves accuracy by ****8.7%**** compared to individual mapping.
- The DT-enabled predictive approach achieves faster convergence: ****63%**** of accuracy after 50% of mission time.

E. Federated Learning Convergence

Fig. 8 shows the FL convergence behavior.

Observations:

- FedAvg with momentum achieves faster convergence than standard FedAvg.
- Non-IID data distribution increases convergence time but FL still converges within 50 rounds.
- Gradient sparsification reduces communication overhead by ****72%**** with negligible accuracy loss.

F. Digital Twin Impact

G. Scalability Analysis

Fig. 9 shows how the framework scales with increasing numbers of UAVs.

[Figure: Mapping Accuracy Comparison]

Fig. 7: Field mapping accuracy (F1-score) as a function of mission time.

[Figure: FL Convergence]

Fig. 8: Training loss as a function of communication rounds.

Results:

- Energy consumption per UAV decreases with more UAVs (35.6% reduction from $V = 1$ to $V = 10$).
- Mission time decreases linearly with V .
- Communication overhead scales as $O(V)$ for FL vs. $O(V^2)$ for CENT.

H. Sensitivity Analysis

Key Findings:

- Energy consumption increases with higher γ but utility saturates after $\gamma = 0.7$.
- The optimal λ_1 balances energy and utility around 0.6.
- Lookahead factor $\xi = 0.5$ provides the best IG-Energy trade-off.

X. CONCLUSION

This paper has presented a comprehensive framework for zero-touch smart agriculture, integrating five synergistic innovations: semantic sensing, adaptive sensor activation, information gain path planning, multi-UAV cooperative learning, and Digital Twin integration. The key findings are:

- 1) **Semantic sensing** reduces communication payload by up to 95% with minimal utility loss, enabling energy-efficient communication in bandwidth-constrained agricultural environments.
- 2) **Adaptive sensor activation** through deep reinforcement learning achieves 18.2% energy savings compared to fixed activation, demonstrating the value of context-aware decision-making.
- 3) **Information gain path planning** improves mapping accuracy by 12.3% compared to conventional approaches, guiding UAVs toward regions of highest uncertainty.

TABLE V: Impact of Digital Twin Integration

Metric	No DT	With DT	Improvement
Energy Consumption (Wh)	24.1	20.5	14.9%
Mission Completion Time (min)	45.2	38.7	14.4%
Revisit Rate (%)	18.3	6.2	66.1%
Prediction Accuracy	—	89.7%	—

[Figure: Scalability Analysis]

Fig. 9: Energy consumption and mission time as functions of the number of UAVs.

- 4) **Federated learning** enables privacy-preserving cooperative knowledge discovery, achieving convergence within 50 communication rounds while reducing communication overhead by 72% through gradient sparsification.
- 5) **Digital Twin integration** enables proactive optimization, reducing mission completion time by 14.4% and revisits by 66.1%, representing a significant step toward predictive agricultural orchestration.

A. Limitations and Future Work

While the proposed framework demonstrates significant improvements, several limitations should be acknowledged:

- 1) **Simulation-Based Validation:** The framework has been validated through simulations; real-world field deployment is needed to validate performance in practical conditions.
- 2) **Computational Overhead:** Onboard semantic encoding and local learning require computational resources that may exceed the capabilities of low-cost UAVs.
- 3) **Communication Reliability:** The framework assumes reliable communication between UAVs and the CS; rural agricultural environments may have limited connectivity.

Future work will focus on:

- 1) **Field Deployment:** Collaborating with agricultural partners to deploy and test the framework in real farm settings.
- 2) **Edge Offloading:** Developing hybrid edge-cloud architectures to offload computationally intensive tasks to edge servers.
- 3) **Resilience:** Extending the framework to handle communication failures and dynamic environmental conditions.
- 4) **Federated Transfer Learning:** Enabling the framework to adapt to new fields with minimal fine-tuning.

The proposed framework represents a significant step toward autonomous, zero-touch smart agriculture, where distributed intelligence, semantic communication, and predictive modeling work in concert to optimize resource utilization

[Figure: Sensitivity Analysis]

Fig. 10: Sensitivity of key parameters: (a) Semantic fidelity γ , (b) Energy weight λ_1 , (c) Lookahead factor ξ .

while preserving data privacy. We believe this work provides a solid foundation for the next generation of sustainable, technology-enabled farming practices.

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